Q) What is an array? TN Note No. 4

Ans) An array is a collection of data elements, all of the same type, accessed using a common name which is called the name of the array. The elements of an array are stored in adjacent memory locations.

Example:- int arr[20];

Here, arr is an array of integers ~~with~~ whose size or dimension is 20.

Q) What is structure? How is a structure member accessed?

Ans) A structure is a user-defined data type which is a collection of heterogeneous data i.e. data can be of dissimilar type. The elements of a structure are always stored in adjacent memory locations. The members of a structure are accessed using dot (.) operator.

Example of a structure:-

```
struct student
{    char name[20];
     int roll;
     int age;
};
struct student s1, s2;
printf("\n Enter the name, roll, age of student 1");
scanf("%s %d %d", &s1.name, &s1.roll, &s1.age);
printf("\n Enter the name, roll, age of student 2");
scanf("%s %d %d", &s2.name, &s2.roll, &s2.age);
printf("\n Name of student 1 is %s, roll no. is %d,
        age is %d", s1.name, s1.roll, s1.age);
printf("\n Name of student 2 is %s, roll no is %d,
        age is %d", s2.name, s2.roll, s2.age);
```

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Q) What is Union? How is a union member accessed?

Ans) A union is a user defined data type that allows storing different data types in the same memory location. In union, all members share the same memory location. A union can have many members, but only one member can contain a value at any given time. Unions provide an efficient way of using the same memory location for multiple purposes.

The members of a union are accessed using dot(.) operator.

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Example:-

```
union student
{
    int roll;
    int age;
    char name[20];
};
union student s1;
s1.roll = 11;
printf("\n Roll no. is %d", s1.roll);
s1.age = 19;
printf("\n Age is %d", s1.age);
strcpy (s1.name, "Rahul");
printf("\n Name is %s", s1.name);
```

Output:-

Roll no is 11
Age is 19
Name is Rahul

Here, we have assigned a value to one ~~variable~~ member at a time and displayed it using printf statement.

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Q) Difference between Structure & Union.

Ans)

Structure	Union
1) The keyword struct is used to define a structure.	1) The keyword union is used to define a union.
2) In case of structure, each member has its own memory space.	2) In case of union, all members share the same memory location; i.e. one memory block is used by all the members of the union.
3) The amount of memory required to store a structure variable is the sum of the size of all the members.	3) In case of union, the amount of memory required is always equal to that required by its largest member.

4) Example
struct stud
{
int a;
char b;
float c;
};
Here, the size of the structure stud is 7. which is the sum of the size of all its members. (2 + 1 + 4 = 7)
	4) Example
	union stud
	{
	int a;
	char b;
	float c;
	};
	Here, the size of the union stud is 4 which is the size of its largest member i.e. float c..

5) All the members of a structure can be accessed at a time.	5) Only one member of a union can be accessed at a time.

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Q) Difference between Structure & Array.

Ans)

Structure	Array
1) A structure is a collection of heterogeneous data i.e. the elements of a structure may be of dissimilar type. A structure may contain 10 elements, out of which 4 elements are integers, 3 elements are characters and the remaining 3 elements are of floating point type.	1) An array is a collection of homogeneous data i.e. all the elements of an array must be of the same type. If an array contains 10 elements, then all these elements must be of the same type i.e. either all of them are integers or all of them are characters or all of them are of floating point type, etc.
2) The individual elements of a structure are accessed using dot (.) operator.	2) The individual elements of an array are accessed using a subscript or index.
3) A structure is a user-defined data type.	3) An array is a derived data type.
4) Accessing structure elements takes more time than array.	4) Accessing array elements takes less time than structures.
5) Example & Syntax	5) Example & Syntax

```
struct stud
{
        int roll;
        char name[20];
        int age;
};
struct stud s1, s2, s3;
```

Here, we declare a structure stud which has 2 integer members and one string member.

```
int arr[20];
```

Here, we declare an integer array of name arr with size or dimension of 20.

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